AI that computes with time and is already winning

A new family of learning machines, built for a world that happens as events, beats the best published models on public benchmarks with a fraction of their size.

5 of 5
datasets won on the public EasyTPP event-prediction leaderboard

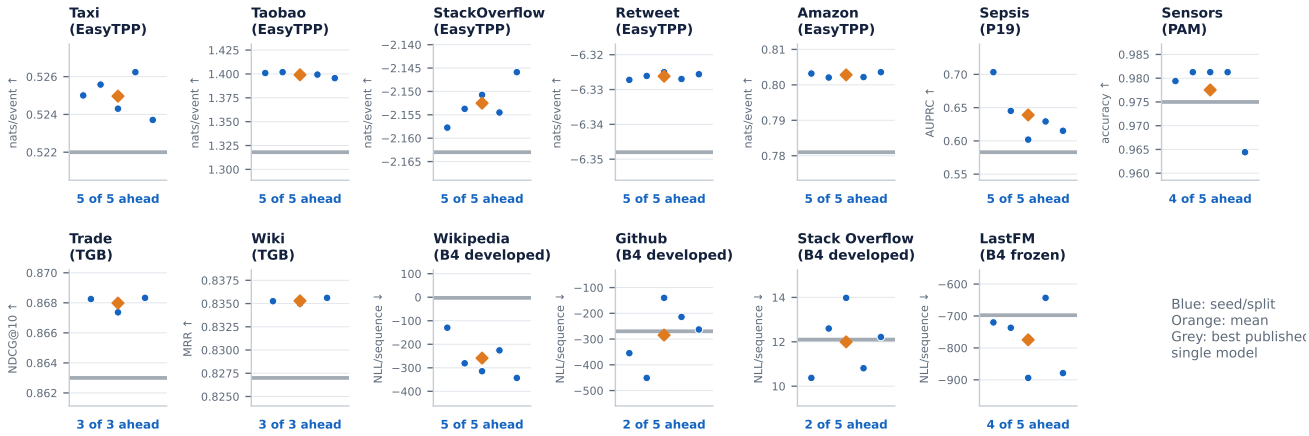
0.639
sepsis early-warning AUPRC vs 0.583 for the best published model ↑

15×
smaller and cheaper than the state-of-the-art model on Retweet, while more accurate

13 wins
public dataset/protocol results ahead of the best published single-model means

In one sentence. Sleeping Machines is building a general-purpose computing substrate for machine intelligence, one that learns through time and sparse events, and its first public results are wins against the strongest published models, achieved by small, efficient models under strict, pre-registered test protocols.

13 public benchmark wins: 53 of 61 seed/split scores ahead



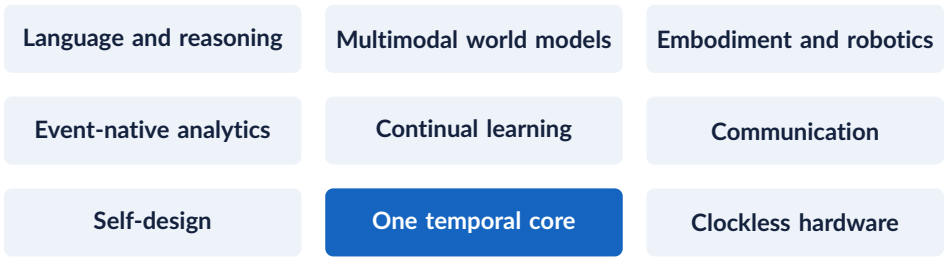
The 13 wins compare dataset/protocol means against the best published single-model result. B4 Wikipedia/Github also beat stronger composite targets; Stack Overflow/LastFM miss those composites. The chart shows individual seeds or official splits, not independent trials. B4 uses recorded-density scoring. The forward-credit development page reports separate private TRAIN-tail experiments, with its scope stated beside the numbers. Each benchmark's unit, and whether higher or lower is better, is stated beside every comparison (↑ higher is better, ↓ lower is better). Sources: `experiments/B1_EASYTPP.md`, `experiments/B2_IRREGULAR_TS.md`, `report/I_SCIENCE.md`.

THE AMBITION

One substrate for machine intelligence

Today's AI runs on a clock: every input is chopped into fixed steps, and the whole model works at every step whether or not anything happened. The real world is not like that. Purchases, heartbeats, sensor readings, messages, machine logs and words arrive as *events*, irregularly, with meaning in their timing and in their silences.

Sleeping Machines builds learning machines in which **time itself does part of the computation**. Our goal is one trainable substrate that spans:



What makes it different, in plain terms:

- **Memories that live through silence.** Each memory keeps evolving between events, so "how long since" and "what happened in between" are information, not missing data.
- **A race of clocks decides what happens next.** Possible next events race against each other; the one that fires first wins. This gives exact predictions of both *when* and *what*, and the same mathematics reproduces the attention mechanism inside today's large language models.
- **Only the relevant parts wake up.** Updates are sparse and addressed: the machine can hold far more capacity than it uses for any single event, which is where the name comes from.
- **Learning from roads not taken.** The model is also credited for the alternatives that lost the race, so it learns which routes to take.

> **Why this matters commercially.** Most of the world's valuable data is timestamped events: transactions, clinical records, telemetry, security and IT logs, user behaviour. Models that understand time natively can be more accurate on that data and much cheaper to run, on CPUs and edge devices as well as in datacentres. The same principles point to hardware without a global clock, where idle capacity costs almost nothing.

THE PROVING GROUND

The benchmarks, in plain language

We compete on established public benchmarks, using their official data splits and scoring rules, against the best results other teams have published. The headline event benchmark is **EasyTPP** (ICLR 2024), the standard test suite for predicting *when* the next event happens and *what* it will be. Its current leader is **S2P2** (NeurIPS 2025, GE HealthCare and UC Irvine), a modern state-space model. The **Temporal Graph Benchmark** (TGB, NeurIPS 2023) tests who interacts with whom, and when, against temporal graph neural networks.

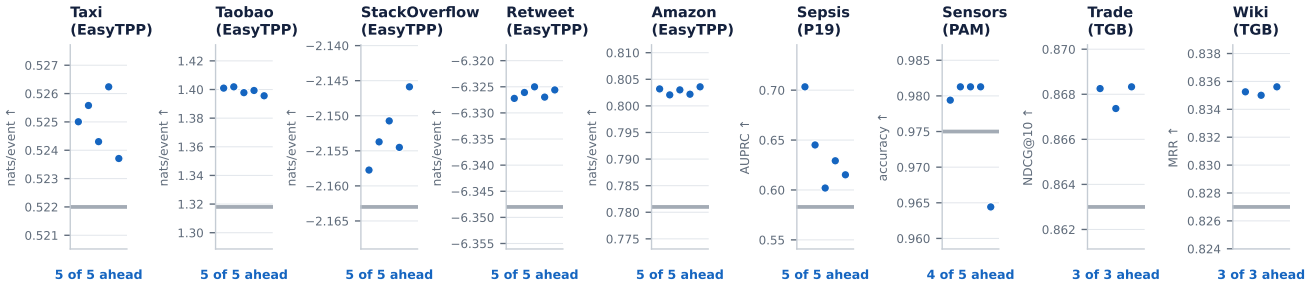
Benchmark	What the data is	Why it matters	Score (unit, direction)
Taobao	Clicks and purchases of users on Alibaba's Taobao marketplace, by product category	E-commerce: anticipating what a shopper does next and when	log-likelihood, nats per event ↑
Taxi	New York taxi pick-ups and drop-offs across city districts	Mobility and logistics: demand and fleet planning	log-likelihood, nats per event ↑
StackOverflow	Badges earned by users of the StackOverflow Q&A site	User engagement and lifecycle prediction	log-likelihood, nats per event ↑
Retweet	Retweet cascades on Twitter, by type of user retweeting	Social media: how information spreads, and how fast	log-likelihood, nats per event ↑
Amazon	Product reviews by Amazon customers, by category	Retail behaviour across product lines	log-likelihood, nats per event ↑
P19 sepsis	PhysioNet 2019: 38,803 intensive-care stays, 34 vital signs and lab values measured at irregular times	Early warning of sepsis, a leading cause of hospital deaths; every hour of earlier treatment matters	AUPRC and AUROC, 0 to 1 ↑
PAM	PAMAP 2 physical-activity monitoring: 5,333 recordings from wearable inertial sensors (17 channels), 8 activities	Wearables and health monitoring: recognising what a person is doing from body-worn sensors	accuracy and F1, 0 to 1 ↑
P12 mortality	PhysioNet 2012: 11,988 intensive-care stays, 36 irregular measurements	Predicting in-hospital mortality from the first 48 hours	AUROC and AUPRC, 0 to 1 ↑
TGB trade	UN trade flows between 255 countries, 1986–2016	Which partners a country trades with most next year	NDCG@10, 0 to 1 ↑
TGB wiki	157,474 timed Wikipedia edits by 8,227 users on 1,000 pages	Which page a user edits next	MRR, 0 to 1 ↑
Neural-TPP benchmark (TMLR 2023/2025)	Seven event datasets (LastFM listening, MOOC course activity, Github, Stack Overflow, Wikipedia edits, MIMIC2 hospital visits, Retweets), five fixed splits each	A second, independent event benchmark: when the next event happens and what it is	negative log-likelihood, nats per sequence ↓

Reading the scores.

- **Log-likelihood (nats per event), higher is better.** It measures how much probability the model gave to what actually happened, both its timing and its type. A gain of 0.08 nats per event means the model assigns about 8% more probability to every real event, on average, compounding over every event in a sequence. Values can be negative; less negative is better.
- **AUPRC, higher is better (0 to 1).** How well the model ranks the rare patients who develop sepsis (4% of stays) ahead of everyone else, with emphasis on raising true alarms without false ones. It is the clinically relevant metric when the condition is rare. **AUROC** (0 to 1, higher is better) measures overall ranking.
- **Negative log-likelihood per sequence, lower is better.** The same probability measure summed over a whole sequence and negated, as the neural-TPP benchmark reports it; time and type ("mark") parts are reported separately and add up to the total.
- **NDCG@10 and MRR, higher is better (0 to 1).** How well the model ranks the true next partners or pages near the top of a list of all candidates.
- **Size and cost, lower is better.** Parameters measure how big a model is; multiply-adds per event measure the work needed for each prediction, which drives running cost and energy.

THE WINS

Ahead of the best published models



Nine established results within the 13-win record; four B4 results follow. Blue dots are official seed/split scores; grey lines are best published single-model means. Each panel states its unit and direction. Official splits are not independent trials.

Benchmark	Best published	Sleeping Machines	Margin	Our size and compute vs the leading model
Taobao nats/event ↑	1.318 (IFTPP)	**1.399 ± 0.003**	+0.081	0.92× S2P2's compute per event; best time and type predictions
Taxi nats/event ↑	0.522 (S2P2)	**0.525 ± 0.001**	+0.003	**1/12 of S2P2's size and compute**; reproduced independently on separate hardware
StackOverflow nats/event ↑	-2.163 (S2P2)	-2.144 ± 0.004	+0.019	1.26× compute; a model of S2P2's size also wins (-2.153)
Retweet nats/event ↑	-6.348 (NHP)	-6.326 ± 0.001	+0.022	**1/15 of S2P2's size and compute**; S2P2 scores -6.365
Amazon nats/event ↑	0.781 (S2P2)	**0.803 ± 0.001**	+0.022	**0.29× S2P2's compute**, 3.6× smaller; best time and type predictions
Wearables, PAM accuracy ↑	0.975 (MTM)	**0.978 ± 0.007**	+0.003	**19× smaller** than MTM (46,316 vs 873K parameters); F1 0.980 vs 0.976; 4 of 5 splits ahead
Sepsis, P19 AUPRC ↑	0.583 (MTM)	**0.639 ± 0.039**	+0.056	62,681 parameters; AUROC 0.916 vs 0.903 ↑; without its temporal memory the same network drops to 0.572 on every split
Trade graphs, TGB tgbn-trade NDCG@10 ↑	0.863 (NAVIS)	**0.868 ± 0.0005**	+0.005	2,107 parameters (NAVIS 1,280); 3 of 3 sealed seeds ahead; a new domain
Wikipedia edits, TGB tgbl-wiki MRR ↑	0.827 (TPNet)	**0.835 ± 0.0003**	+0.008	7,995 parameters; 3 of 3 sealed seeds ahead; 4.5 ms per query on one CPU thread

± is the standard deviation across the runs (five; three on TGB). Compute is multiply-adds per event, counted from S2P2's released code at its published configuration.

One model for the whole benchmark. A single configuration of our model, with no per-dataset tuning, is ahead of the best published result on all five EasyTPP datasets: Taobao 1.397 vs 1.318, Taxi 0.526 vs 0.522, StackOverflow -2.144 vs -2.163, Retweet -6.324 vs -6.348, Amazon 0.802 vs 0.781 (nats per event ↑; 24 of 25 runs ahead; pre-registered protocol), at 0.07× to 1.03× the leading model's compute. Across all 13 winning dataset/protocol entries, 53 of 61 seed/split scores are ahead (B4 results on the next page).

BEYOND EASYTPP

Four wins on a second event benchmark

One model configuration won all five EasyTPP datasets. Applied unchanged to a second event benchmark, it also beats the best published single-model total on **Github activity and LastFM listening**. These frozen-transfer results use the independent neural temporal-point-process benchmark of Bosser & Ben Taieb (TMLR 2023, 2025). Developed pointer models now win on **Wikipedia and Github** against composite targets stronger than any single published model; a developed Stack Overflow model also beats the best single published total.

Total prediction loss nats/sequence ↓	Published reference	Sleeping Machines	Improvement
Wikipedia, developed	-122.62 (composite target)	-258.76	**136.14**
Github, developed	-272.90 (composite target)	-284.47	**11.57**
Stack Overflow, developed	12.10 (LNM++)	12.00	**0.10**
LastFM listening	-697.50 (LNM++)	-774.64	**77.14**

Five official splits; mean total negative log-likelihood per sequence, lower is better. Standard errors across the five splits: Wikipedia developed 37.73; Github developed 54.29; Stack Overflow 0.65; LastFM 48.38. Frozen Github: -270.62 (SE 48.69).

Why this matters. Github and LastFM demonstrate transfer of a common event model. Developed Wikipedia and Github now beat the stronger composite targets, after validation-only variant selection.

Stronger timing predictions and addressed memory

The component scores show where the temporal architecture contributes:

Component nats/sequence ↓	Best published	Sleeping Machines	Achievement
LastFM: event timing	-1363.78	-1380.05	16.27 lower timing loss
Github: event timing	-382.40	-390.93	8.53 lower timing loss
Stack Overflow: event timing	-91.10	-91.60	0.50 lower timing loss
Wikipedia, frozen: event type	144.79	**27.65**	80.9% lower type loss

Memory that copies what it has seen. About 90% of consecutive Wikipedia edits return to the page just edited, while 10–27% of test pages never appear in training. Our addressed memory stores pages as they occur and can reuse them later in the sequence. Its type-prediction loss is **27.65 versus 144.79** for the strongest published type predictor. The developed pointer model now wins on total loss: -258.76 **versus the** -122.62 **composite target**, with type loss 18.19. All five official splits beat the target; validation selected the variant, then TEST was scored once. The original frozen-transfer failure motivated the stability repair.

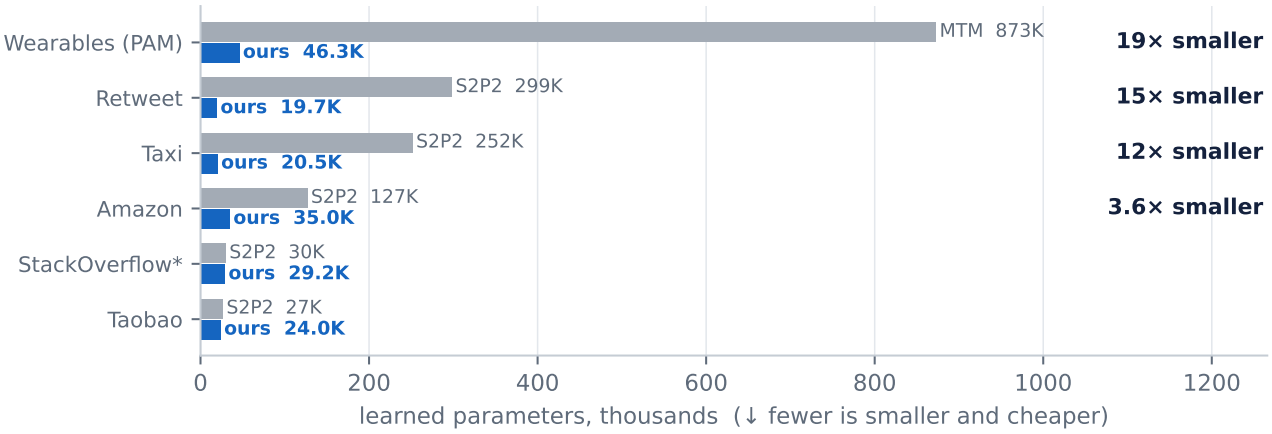
Published models and a separate stretch target

The pre-registered research target combines the best timing score and best type score from *different* published models. No single published model reaches that composite. Frozen Github and LastFM beat the strongest single-model totals while missing the stricter composite targets (-272.90 and -849.65). Developed Github and Wikipedia now beat their composites. Stack Overflow beats the best single model (11.9956 vs 12.10), while missing the 11.90 composite. MOOC, MIMIC2 and Retweets need total-score wins.

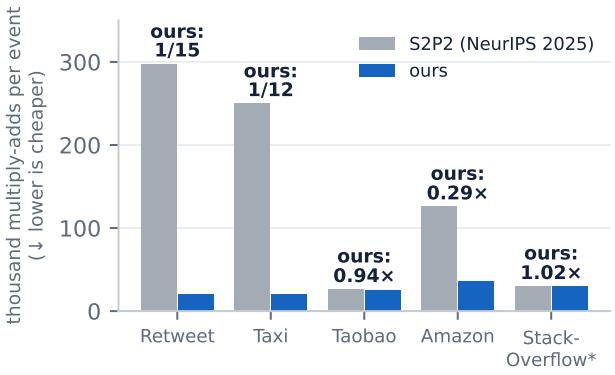
Protocol and complete record. Validation selected checkpoints; five fixed test splits; numerical guarded reruns as specified in B4. Github has 1.3% zero gaps: published log-normal references replace zero by 10^{-20}; our timing model floors at the recording cell. The comparisons above retain those recorded scoring conventions; the developed Wikipedia result is a recorded-density win, not a dequantization or interval-score result. Full seven-dataset totals, numerical failures, rerun rules and recording-grid audits: `experiments/B4_NTPP_BENCHMARK_PROPOSAL.md` and the report's B4 loss analysis. Developed result: `investment/b4_developed_evidence.json`.

EFFICIENCY

Winning while small



Learned parameters, in thousands; fewer means a smaller, cheaper model (lower is better). Grey: S2P2, the leading published state-space model on EasyTPP, at its published configuration for each dataset; for PAM, MTM (2025) at the parameter count it reports for PAM. Every Sleeping Machines model shown is also more accurate than the model beside it. *StackOverflow: the model matched to S2P2's size, which also wins.



Work per prediction, in thousands of multiply-adds per event (lower is cheaper). "ours: 1/15" means our model needs one fifteenth of S2P2's work.

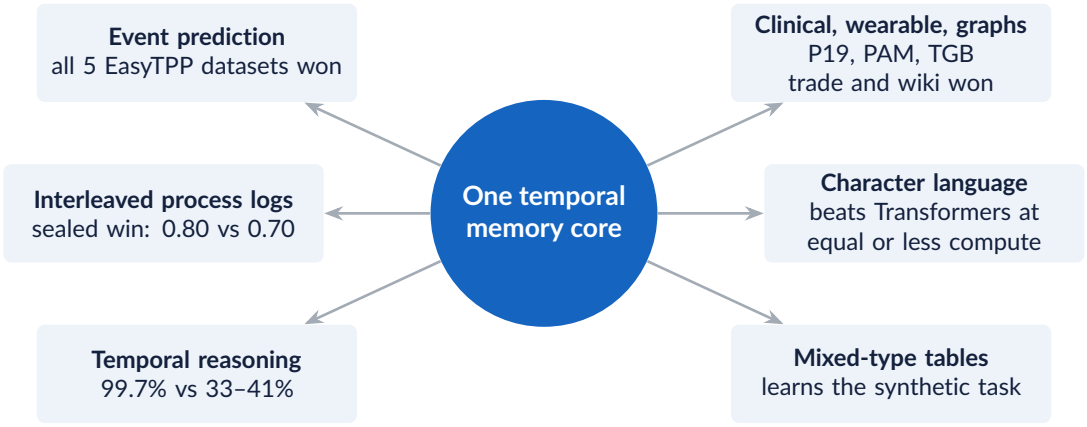
What size buys.

- **Running cost and energy** scale with the work per prediction.
- **Deployment anywhere**: models of 20,000 to 60,000 parameters run on ordinary CPUs, phones and embedded devices, close to the data.
- **Privacy**: clinical and industrial data can stay on site.
- **Headroom**: accuracy at small size means the architecture, not brute scale, is doing the work.

GENERALITY

One family across different kinds of data

S2P2 is shown on one kind of task. Our family wins on four kinds of data (event streams, clinical records, wearable sensors, interaction graphs) with shared machinery: the sepsis classifier reuses *the same temporal memory layer, same code and same size*, as the event-prediction model.



One shared core across four domains (development data, first seed). We trained a *single* temporal memory of 14,080 parameters on taxi rides, online shopping, Q&A activity and product reviews at the same time; only the small input and output layers differ per dataset. It matched or beat the four separately trained models on three of the four (log-likelihood, nats per event, higher is better: Taxi 0.490 vs 0.487, Taobao 1.290 vs 1.284, StackOverflow -2.186 vs -2.177), with about 40% fewer parameters in total. Mobility, shopping and online-community timing share one learned temporal dynamics. Amazon trailed and was still improving when training stopped; per-dataset training budgets are the next test.

Why generality matters for value. A company with one specialised model has one product. A company with a core that transfers across data types can serve many markets from one technology, and every new domain makes the core stronger.

WHY THE LARGER VISION IS ACHIEVABLE

Each layer of evidence supports the next

Next: scale and transfer. Large-scale language on GPUs, skill transfer between data types, clockless hardware.

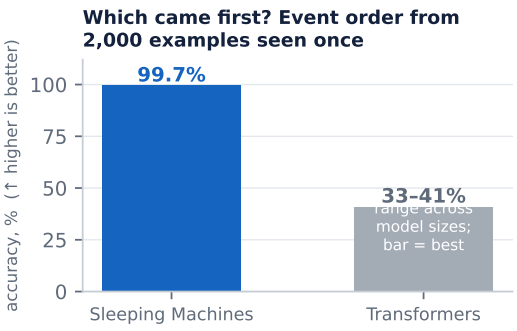
5. It generalises. One shared core across four event domains; one memory layer for events and clinical records.

4. It wins public benchmarks. 13 public benchmark wins: five EasyTPP, P19, PAM, two TGB and four Bosser–Ben Taieb results; 53 of 61 seed/split scores ahead. Models up to 15× smaller on the established event tasks.

3. It learns language efficiently. On 10M characters it beats a Transformer: 1.888 vs 1.908 bits per character ↓ with 0.40× its training compute. At 90M characters it beats a tuned Transformer-256×4 (1.783 vs 1.811 ↓) at 1.06× its compute.

2. The mechanisms work. Event order learned from 2,000 examples seen once: 99.7% accuracy vs 33–41% for Transformers ↑. Timing-only discrimination: 95.3% where any order-only model is capped at 50% ↑. Learning from roads not taken improves language 2.507 → 2.371 bits per character ↓ for 0.3% extra training work; more idle capacity improves it further at the same active work.

1. Mathematics: it contains today's leading architectures. A race of clocks reproduces the attention of Transformers *exactly*, and Mamba-style selective state-space models are exact members of the family (verified numerically to 4×10^{-13}). Everything those models can represent, ours can, plus native time and sparsity.



Why the steps add up. The family is not a narrow trick for one benchmark. Its mathematics contains the architectures that power today's AI; its mechanisms solve temporal problems those architectures fail at; it already learns language competitively at small scale; and it wins public benchmarks on four kinds of data with one shared memory design. Scale is the remaining work, and each layer beneath it is already demonstrated.

Hardware. In our cost model, the native language model moves 5.8× fewer bytes per character than a comparable Transformer at better quality (modelled, not yet measured on silicon; a funded milestone).

RIGOR

How we know the results are real

Sealed tests, rules fixed in advance. Every win uses the benchmark's official data splits. The reporting rule is written down before the test set is touched, and each model is scored on the test set exactly once.

The benchmark's own scorer agrees. Re-scoring our models with EasyTPP's own estimation method gives the same results within its sampling noise.

Independent runs. Benchmark headline results are averages of independent runs (five seeds or official splits; three pre-registered seeds on TGB), with their spread: 53 of 61 seed/split scores are ahead across the 13 winning entries. Split scores are not independent trials; wins compare dataset means.

We found and closed a scoring pitfall. Timestamps in these datasets are rounded (for example to whole seconds). Flexible models can exploit the rounding for large, fake gains. We built an audit that removes the rounding; we withdrew our own early results that failed it, and every reported win passes it.

Reproduced on separate hardware. The Taxi win was reproduced independently on a separate desktop computer with separately downloaded data: 0.5252 vs 0.5250 nats per event ↑, every run above S2P2. One-command kits let third parties rerun the EasyTPP and both TGB wins.

Losses reported as plainly as wins. Where we are behind, we say so with the numbers and the next test (next page). Comparisons always use the same units and protocol as the published result.

LEARNING MORE ECONOMICALLY

Teaching the model how to learn

The earlier pages show that our models can make strong predictions with small models. We are also tackling the cost of *learning*: deciding which parts of a model should change when new information arrives.

A sparse model takes only a few routes through its available memory and computation. To improve, it needs to learn from those choices, including alternatives it did not take. Trying every alternative can make training expensive even when prediction is cheap.

> **Our approach.** Use what the model discovers while making predictions to help it evaluate alternatives and choose better learning updates. The ambition is a model that learns both *what to predict* and *how to improve*, using the same event-based architecture.

98.4%
less error when estimating the value of alternative learning routes

2.09×
faster evaluation of alternatives in the measured experiment

77.8%
less randomness in the route controller's learning signal

What these results mean.

- **More accurate guidance.** On eight new language samples, information from the model and its learning state estimated the relative value of alternative routes more accurately than a fixed estimate. Every sample improved.
- **Less repeated work.** Reusing this information, with an exact correction, took 12.9 seconds instead of 26.9 seconds for separately evaluating all alternatives. Both measurements include discovering the alternatives.
- **A steadier learning signal.** Random route selection introduces noise into learning. In a separate controller experiment, the corrected signal reduced that source of noise while preserving its average direction.

Significance. These experiments demonstrate that information already produced by a model can make an expensive part of learning more accurate and more economical. They give a concrete engineering basis for our learning architecture. The measured speedup applies to evaluating learning alternatives, not to training an entire model.

Learning rules can themselves improve. In a separate event-prediction experiment, a trained controller adjusted the model's learning updates and improved prediction quality in all nine tested combinations of model depth and initialization. Those fits used more CPU time. This shows that learned update rules can improve quality within our model family.

Status: a working research prototype. The 98.4% result uses eight private language samples; speed uses four samples and three timing repeats; noise uses four samples and a fixed controller. These are separate component measurements. The update-controller result uses three initializations at depths 2/4/8 on Taxi training/development data and retains ordinary error propagation. Detailed methods and completed result hashes are in `investment/forward_credit_evidence_20261011.json`.

ONLINE MARKET PREDICTION WINS

Confirmed on a new month, chosen in advance

Our native model beats both Transformer and LSTM on chronological market forecasts while **all models adapt their weights online**. Each forecast is scored before its outcome is used for learning. The win held on **1 October data that no model had seen**, with every choice fixed and published from August evidence before the October data were acquired.

Lower prediction loss is better. 1 October: same 65,536 consecutive scored trades.

Model	Prediction loss	Online GFLOPs
Sleeping Machines	**8.07854**	**8.51**
LSTM	8.10271	9.51
Transformer (cached)	8.17919	124.68
Transformer (re-encoded)	8.17638	7,963.12

Native wins by 0.02417 nats/event over LSTM and 0.10066 over cached Transformer, and leads in all 16 of 16 consecutive 4,096-trade windows. The scored forecasting-and-learning updates use **14.7 times less arithmetic than the cached Transformer** and 936 times less than the re-encoded one.

Why this matters. Markets change; a forecaster has to keep learning while it predicts. This test carries each model's complete August learner, memory and optimizer state across a five-week calendar gap and lets it adapt on the new month. The ranking and the work advantage both transfer. On the August development stream (65,536 forecasts) the same models score 5.17588 native, 5.19672 LSTM, 5.27966 re-encoded and 5.28043 cached Transformer. Stored weights plus numerical prediction state require **0.146 MB versus 2.334 MB** for the cached Transformer (16 times less, logical tensor bytes at reference precision).

What each model is allowed to do. All receive the same chronological observations and update after 32 revealed outcomes. Native temporal state and LSTM recurrent state persist. The Transformer keeps 2,048 events of history, either as cached past-version keys/values or re-encoded with current weights after every update, plus persistent addressed output memory. No memory reset is imposed at update boundaries or across the calendar gap.

Three-seed acquisition complete. Native mean August development loss is **6.8842**, versus LSTM **6.8996** and Transformer **6.9686**; native leads both at every paired seed. These fixed-weight results are separate from the single-seed October online win. Next: reuse saved learners for online replication with unchanged settings. October is already accessed; the seed extension is replication, not a new prospective test. September remains sealed.

Scope: two-layer, width 32 models; seed 0; Binance BTCUSDT public trades. Acquisition on 24 August TRAIN; August development stream 25 August. October: preregistration committed before data acquisition, 128 context events then 65,536 forecasts, no October tuning. Online arithmetic covers scored-stream forward/backward/optimizer work; original fitting, warmup, context-only preparation and gates are separate; special functions separate. CPU wall time is an implementation diagnostic. No trading-return or measured chip-energy claim. Source-bound results and shared units:
`investment/online_market_evidence.json`.

INDUSTRIAL FAULT DETECTION

A sealed win on interleaved factory logs

A factory log interleaves the events of many items and machines with no item labels. A failing machine shows up as steps that slowly take longer. Our model learns, from normal logs alone, how long each production step takes, links each event to the earlier event it most likely continues, and keeps a separate running anomaly score for every step. **A slowing machine stands out within the first half of a production run.**

Higher is better. Fault detection (AUROC) after 512 events per line, 2,000 normal and 2,000 faulty runs.

Detector	AUROC
Sleeping Machines (three independent fits)	**0.801**
Time-encoded Transformer (three seeds)	0.704
Best of nine classical detectors	0.676
LSTM (three seeds)	0.667

Ahead by 0.097 AUROC (95% interval 0.088 to 0.121); every one of our three fits beats every reference. Wear faults 0.772 versus 0.650; retry faults 0.829 versus 0.758.

Why this matters. Predictive maintenance runs on exactly this kind of data: anonymous, interleaved machine events where timing carries the signal. The model binds events through time, the family's core mechanism: delayed clocks racing to explain each event.

How the test was run. The test data were generated by our open simulator only after the full plan (model, settings, references, decision rule) was committed. Each model was scored once. No item identities are used anywhere; the model's knowledge is fixed during scoring, so it cannot learn a fault as normal. An identity-aware oracle, not eligible, reaches 0.821 on the earlier test.

Scope: FAS v2 simulator (Keski-Valkama 2017), two interleaved assembly lines of 30 items, 2% dropped events; references are the pre-selected Transformer and LSTM configurations of the earlier sealed test plus nine information-matched classical detectors. Source-bound result: `investment/fas_evidence.json`.

LANGUAGE

Addressed memory leads our language models

Our strongest language models give the network explicit, addressed access to its own past: events write keyed messages, and later tokens read the relevant ones directly. This is the same principle behind the fault-detection win on the previous page.

Lower is better. Prediction loss, nats per token, on the same 8,192 FineWeb development tokens (GPT-2 tokens, 1,024-token history).

Model	Prediction loss
Causal event race model (one pass over 4M tokens)	**5.4665**
Keyed temporal-memory model (two seeds, three passes)	5.60 / 5.62
Dense hybrid diagnostic, eight layers (one pass over 4M tokens)	5.8414
Earlier flagship + addressed episodic memory	5.634
Earlier flagship (without addressed memory)	5.978

Addressed memory is worth 0.35 nats per token on the flagship itself; removing the keyed read from the keyed model costs 0.10 at 1M training tokens.

Learning more efficiently. Sparse counterfactual teaching improves prediction by **0.093 nats per token** while cutting complete CPU training time by **15.3%** (6.528 vs 6.621, 7,734 vs 9,129 seconds at 8.39 million tokens). Calibrated local learning controls, where each unit limits its own update without a global step, add a further **0.005 nats** at **+0.41%** CPU time. Local control removes one chip-wide synchronization requirement: the basis for clockless hardware, where independent units run at their own pace (ETH Zürich; DTU). No chip speed or energy gain is claimed yet.

Depth that earns its work. A dense keyed hybrid, now a labelled diagnostic control (founder direction: build our own models), improves development loss from **5.9572 to 5.8987 to 5.8741** at two, four and eight layers on the same 65,528-target slice and 1.05M-token corpus (one seed; five/six/six epochs, different total work). Completed 16-layer loss is 5.8879; two asynchronous hierarchy variants score 5.9392 and 5.8911. Eight layers remain selected. This protocol differs from the 8,192-token table above. In one pass over 4M fresh tokens, eight layers also beat two: **5.8414 versus 5.8648** on that flagship development protocol (equal updates, different parameters/work).

Our own race model beats the dense hybrid. The causal event race model predicts each token through a race among its candidate causes: copying or continuing earlier text, dormant concept units woken by the context, and a background word race. On the same 4M fresh FineWeb tokens, order and schedule it scores **5.4665** on the flagship development protocol against **5.8414** for the eight-layer dense hybrid, and **5.3094** against 5.6955 on the 65,528-target slice: 0.38–0.39 nats better while using 3.3 times less training compute per token (7.8M versus 25.9M FLOPs, counting the exploration term and the optimizer) and 3.2 times less CPU time. Its race causes deliver 0.45 nats beyond its background route. One seed, development slices; two temporal-memory state layers and one race level, with the compositional hierarchy of races next. Source: `experiments/results/token_language/curie_r1_cerl_4M_s0_20261016_v1.json`.

Fully native learning. Eight native layers complete 65,536 chronological training targets: selection NLL improves from **7.9307 to 6.9981**; final flagship development NLL is **7.0473** (8,192 targets, 1,024-token windows, one seed). The acquired weights, Adam state and persistent history continue toward 4M cumulative targets with lower-memory queries, joint composition and delayed local producer credit. This member uses no recurrent BPTT or SSM/attention/MLP carrier; its optimizer is synchronous. Learned document-relevance biases pass normalization, gradient and recovery contracts. The fully native deep milestone is still in development. Source: `investment/local_native_language_evidence.json`.

The next scaling proof: learning compositions. Higher stages query several persistent lower memories and learn joint refinements. Delayed task credit should teach the actual memory producers, so useful parts can learn to cooperate. Bounded child-credit replay is implemented; its bounded gradient, unchanged-prediction, work-limit and acquired-learner recovery gate passes; the full-depth sampled extension awaits acquired-model validation. The investment milestone is improved held-out prediction and reusable relations from deeper refinement at disclosed total learning work, followed by useful capacity with measured activity, discovery and traffic. These are next proofs, not completed scaling results. The two-page companion, `investment/sleeping_machines_composition_demonstration.pdf`, connects the completed language result to this architecture and proof plan.

Native language composition at 16M tokens. Four-level residual memory scores **5.0433 NLL**, improving the native causal-race member's **5.0634**; the saved sixteen-layer dense hybrid scores 5.2034. One seed, one pass over

16M FineWeb GPT-2 tokens; 8,192 development targets in 1,024-token windows. Current-slot memory composition, with old-slot rewrites disabled. This is a development accuracy result; fully asynchronous deep learning and complete learning arithmetic are separate tests. Source: `investment/native_composition_16m_evidence.json`.

Old-memory refinement makes depth pay more. When the four residual-memory levels may also write corrections into older memory slots (later context refining earlier memories, which a Transformer's frozen key/value cache cannot do), the same 16M-token run scores **4.9901** on the flagship development protocol: 0.073 nats better than the single-level race model (5.0634) and 0.21 better than the sixteen-layer dense hybrid (5.2034; eight layers 5.2183, four layers 5.2699), with 15.5M versus 32.2M training FLOPs per token. On the 65,528-target slice, whose lanes are 8,192 tokens long against 256 in training, it is worse (5.0874 versus 5.0282): corrections accumulate over 32 times more history than training showed; a learned decay of old corrections is the fix under test. One seed. Source: `experiments/results/token_language/curie_r1_rmr_resid_K4_16M_s0_20261016_v1.json`.

Memories compose, then emit. On one 32-event fixture, native views reduce messages **349 to 125** and candidate scores **11,168 to 4,000**; timer work rises from 2 to 188 scores. These are mechanism/work results, not equal-quality or energy wins. Observed relative frames eliminate pure-translation drift. The acquired role-memory actor scores **0.68661 DEV loss**, versus 0.71641 for a carried-view extension and a 0.69315 prior; neither clears the transfer criterion. Grounding reduces root-geometry error by **34.6%** without improving class transfer. Hierarchy TEST is not model-scored.

Wholes keep their parts accessible. A higher memory now retains references to the exact child-memory versions it bound, so later lower-memory changes do not rewrite an existing whole. The acquired eight-level, 32-stage actor preserves outputs, completion times and parameter gradients bit-for-bit, with no added actor parameters. An auxiliary native race learner pools both root parts with live lower-memory credit. **16 numerical checks** pass. This is an implemented information path; stable semantic identities and relation transfer are the next test.

Composition test: more training did not teach whole-object structure. A repair of how pixels are grouped into parts fixed a defect affecting 25% of groups (all eight parts now complete; 19 checks pass). On the synthetic task, the native hierarchy must tell apart objects built from the same parts in different arrangements. It recognises every part (512 of 512) but still classifies whole objects at chance after 32 times more training: 50% accuracy and NLL 0.693, the value of a coin flip, on held-out and fresh scenes. It does not fit the training arrangements either (53%), so the limit is how parts are bound into wholes, not data or training length; the next construction targets that binding step. Source: `investment/composition_scale_evidence.json`.



Bound versions stay readable while live lower memories evolve.

Scope: keyed model 3.33M parameters, repeated epochs over 4M training tokens; flagship and episodic arm single seed; quality placement, not matched compute. Source-bound results: `investment/language_memory_evidence.json`, `investment/language_learning_evidence.json`, `investment/locality_scaling_evidence.json`, `investment/hierarchy_development_evidence.json`, `investment/bound_hierarchy_evidence.json`

WHAT THE EVIDENCE ADDS UP TO

A proven prediction engine and a developing learning engine

Sleeping Machines aims to be a general-purpose substrate for language and reasoning, multimodal world models, embodiment, continual learning, communication, self-design and hardware. Its evidence already spans several parts of that ambition.

Strong prediction with small models. The public benchmark results lead this brief because they are the strongest evidence: 13 public benchmark wins across the five EasyTPP datasets, clinical and wearable tasks, two graph protocols and four results on a second event benchmark, with models up to 15 times smaller. Two preregistered confirmations add a new month of market forecasting while every model learns, and a sealed industrial fault-detection test: 0.801 versus 0.704 AUROC for the best Transformer. That gives the model family a measured advantage on specific tasks, using existing computers.

A practical route to improving learning. We have measured more accurate and faster evaluation of learning alternatives, better predictions from learned update rules, and online market learning that stays ahead of LSTM and Transformer learners on unseen data. Together they show working pieces of a machine that learns how to learn. Each result has its own scope; the pieces are not one combined speedup.

A mathematical foundation. We have derived and numerically checked how route-level learning errors depend on prediction uncertainty and differences between alternatives. The direct route formulation can avoid multiplying local estimation errors through successive layers. This explains why the design is worth pursuing; it is an analytical result, separate from measured scaling.

Where we are behind. On P12 mortality the family scores 0.871 versus 0.880 AUROC. On the second event benchmark (page 5), our current model with addressed episodic memory is in development; its verdicts follow when the test splits are scored.

Where scaling stands. Small-model event wins and keyed-memory language progress support scaling the family. The strongest token result in this brief is 5.60 nats after 4M training tokens; a separate fixed-data keyed construction improves with two, four and eight layers. Native composition now separates persistent state from emitted work and controls coordinate nuisance, but two short unseen-combination development fits remain at 50% accuracy. Reusable part representations and useful credit across their composition are the next proof; adding depth alone will not supply them.

The commercial significance. The public wins support the value of small, accurate models for timestamped data in industry, IT operations, security and finance; the sealed fault-detection win and the new-month market win show it on predictive maintenance and live market data. The learning work addresses a further source of cost: keeping models useful as new information arrives. If the measured component advantages carry through to complete applications, they could make continual adaptation more affordable on local machines and support sparse, clockless learning hardware. That broader opportunity is the significance of the design, rather than a measured revenue, energy or whole-model training claim.